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சமநிலையில்

$B \rightleftharpoons A$

95.2% 1.3% $\Rightarrow$

$K_1 = \frac{1.3}{95.2} = 0.0136$

$B \rightleftharpoons C$

95.2% 3.5% $\Rightarrow$

$K_2 = \frac{3.5}{95.2} = 0.0367$

$\Rightarrow \Delta G_1^0 = -2.303\ RT\log K_1$

$\Delta G_1^0 = -2.303 \times 8.314 \times 448 \times \log 0.0136$

$\Delta G_1^0 = +16010\ J$

$\Delta G_1^0 = +16\ kJ$

$\Rightarrow \Delta G_2^0 = -2.303\ RT \log K_2$

$\Delta G_2^0 = -2.303 \times 8.314 \times 448 \times \log 0.0367$

$\Delta G_2^0 = +12312\ J$

$\Delta G_2^0 = +12.312\ kJ$

67.

தீர்வு

கொடுக்கப்பட்டவை

$T = 33K$

$N_2O_4 \rightleftharpoons 2NO_2$

ஆரம்பச் செறிவு	100%	0
பிரிகையடைந்த செறிவு	50%	-
சமநிலையிலுள்ள செறிவு	50%	100%

$K_{eq} = \frac{100}{50} = 2$

$\Delta G^0 = -2.303\ RT\log K_{eq}$

$\Delta G^0 = -2.303 \times 8.314 \times 33 \times \log 2$

$\Delta G^0 = -190.18\ J\ mol^{-1}$

68.

தீர்வு

கொடுக்கப்பட்டவை

$\Delta H_f^0\ (SO_2) = -297\ kJ\ mol^{-1}$

$\Delta H_f^0\ (SO_3) = -396\ kJ\ mol^{-1}$

$SO_2 + \frac{1}{2} O_2 \rightarrow SO_3 \quad \Delta H_r^0 = ?$

$\Delta H_r^0 = (\Delta H_f^0)_{compound} - \Sigma (\Delta H_f)_{elements}$

$\Delta H_r^0 = \Delta H_f^0\ (SO_3) - \left(\Delta H_f^0\ (SO_2) + \frac{1}{2} \Delta H_f^0\ (O_2) \right)$

$\Delta H_r^0 = -396\ kJ\ mol^{-1} - (-297\ kJ\ mol^{-1} + 0)$

$\Delta H_r^0 = -396\ kJ\ mol^{-1} + 297$

$\Delta H_r^0 = -99\ kJ\ mol^{-1}$

69.

தீர்வு

கொடுக்கப்பட்டவை

$T = 298\ K$

$\Delta H = 400\ J\ mol^{-1} = 400\ J\ mol^{-1}$

$\Delta S = 0.2\ J\ K^{-1}\ mol^{-1}$

$\Delta G = \Delta H - T\Delta S$

if $T = 2000\ K$

$\Delta G = 400 - (0.2 \times 2000) = 0$

if $T > 2000\ K$

ΔG will be negative.

The reaction would be spontaneous only beyond 2000K